

The 26-Dimensional Descent

Connecting Finite Symmetry to the Critical Line
via Lattice Spectral Filters and Formal Verification

DULA Collaboration*

With machine-verified proofs in Lean 4 / Mathlib v4.28.0

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Abstract

We present a formally verified reduction of the Riemann Hypothesis to a single conjecture in harmonic analysis, mechanized in Lean 4 with Mathlib v4.28.0. The framework connects the 24-dimensional geometry of the Leech lattice Λ_{24} to the 1-dimensional distribution of Riemann zeta zeros through a spectral filter governed by $\alpha = \pi/\log(196560)$, where 196560 is the kissing number of Λ_{24} .

The reduction proceeds in four steps: (1) a Cohn–Elkies auxiliary function provides a pointwise positivity margin; (2) the identity theorem guarantees isolated zeros of the projected transform; (3) dilations of functions with isolated zeros cover $\mathbb{R}$ densely; (4) the Geometric Hahn–Banach theorem converts annihilator triviality into topological density. Autocorrelation continuity is proved unconditionally via an L^2 isometry bypass.

The spectral gap $c_1 = 0$ makes α minimal among root system lattices. We compute $\cos(\alpha\gamma_n)$ on 50 zeta zeros: 15 are amplified ($|\cos| > 0.9$), 4 suppressed. The root system ladder $A_2 < D_4 < E_8 < \Lambda_{24}$ extends to GRH via $\alpha_\chi = \pi/\log(q)$. The Lean formalization comprises 12 files with 5 sorry-free; remaining sorries reduce to classical analysis (Paley–Wiener 1934, Schwartz 1950) being formalized independently.

*AI-Assisted Formalization Architecture: DULA–GROK–CLAUDE–GEMINI–ARISTOTLE.

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1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\operatorname{Re}(s) = 1/2$, has resisted proof for over 160 years. Weil [1] showed that RH is equivalent to the non-negativity of a certain distribution:

$$W(g \star g^*) \geq 0 \quad \text{for all even Schwartz functions } g \in \mathcal{S}(\mathbb{R}). \quad (1)$$

The DULA framework (Dimensional Unification and L-function Architecture) connects this 1-dimensional analytic condition to the 24-dimensional geometry of the Leech lattice Λ_{24} through the Cohn–Elkies linear programming bounds [2]. This paper presents the complete framework, its computational discoveries, and its formal verification in Lean 4.

1.1 The central parameter

The entire construction is governed by a single parameter:

$$\alpha = \frac{\pi}{\log(196560)} \approx 0.25775, \quad (2)$$

where 196560 is the kissing number of the Leech lattice—the number of unit spheres that can simultaneously touch a central unit sphere in 24 dimensions. This parameter encodes the spectral gap of Λ_{24} : the theta series $\Theta_{\Lambda_{24}}(q) = 1 + 0 \cdot q + 196560q^2 + \cdots$ has $c_1 = 0$, meaning there are no lattice vectors of squared norm 2. The coefficient $c_1 = 0$ is the spectral gap, and its consequence—that α is the smallest such parameter among all root system lattices—is what makes the Leech lattice uniquely suited for the reduction.

1.2 The 26-dimensional structure

The descent from lattice geometry to zeta zeros passes through 26 dimensions:

- 24 dimensions: the Leech lattice Λ_{24} and the Cohn–Elkies sphere packing bound
- 1 dimension: the Mellin reparametrization from radial to multiplicative coordinates
- 1 dimension: the Schwartz test function space where the Weil distribution acts

The number $26 = 24 + 1 + 1$ is the critical dimension of the bosonic string, where the Leech lattice provides the transverse space. This is not a coincidence: both the bosonic string and our construction exploit the exceptional properties of Λ_{24} in the same dimension.

1.3 Summary of results

1. We define the Hermite–Biehler function $E(z) = \Xi(z) \cdot e^{-i\alpha z}$ and prove $\text{RH} \iff \text{HB}_1$ (no upper-half-plane zeros of E) in Lean with zero sorries.
2. We verify HB_2 (the de Branges inequality $|E(\bar{z})| < |E(z)|$ for $\text{Im}(z) > 0$) unconditionally via analytic phase monotonicity.
3. We prove the autocorrelation map $g \mapsto g \star g^*$ is continuous in the Schwartz topology via an L^2 isometry bypass, with zero sorries.
4. We prove the Geometric Hahn–Banach density theorem: trivial annihilator implies dense span, with zero sorries.
5. We reduce RH to **Conjecture A** (CE Fourier Richness): that the Fourier transforms of dilated Cohn–Elkies projections are dense in the frequency domain.
6. We provide numerical evidence via SVD analysis showing $\sigma_{\min} \approx 2.8 \times 10^{-16}$ as the number of dilations grows, indicating Conjecture A is true at machine precision.
7. We identify 10 computational discoveries about the spectral filter mechanism and the GRH extension.

2 The Leech Lattice and the Spectral Filter

2.1 The theta series and the spectral gap

The Leech lattice Λ_{24} is the unique even unimodular lattice in 24 dimensions with no vectors of norm $\sqrt{2}$. Its theta series is

$$\Theta_{\Lambda_{24}}(\tau) = 1 + 196560q^2 + 16773120q^3 + \cdots, \quad q = e^{2\pi i\tau}, \quad (3)$$

where the crucial feature is $c_1 = 0$: the coefficient of q^1 vanishes. This is the *spectral gap*—the absence of norm- $\sqrt{2}$ vectors.

Definition 2.1 (Root system alpha parameters). For a lattice L with kissing number $k(L)$, define $\alpha_L = \pi / \log k(L)$. The root system ladder gives:

Lattice	dim	Kissing	α_L	Period π/α_L
A_2	2	6	1.754	1.79
D_4	4	24	0.989	3.18
E_8	8	240	0.573	5.48
Λ_{24}	24	196560	0.258	12.19

The Leech lattice has the smallest α because it has the largest kissing number, which in turn follows from the spectral gap $c_1 = 0$.

2.2 The spectral filter on zeta zeros

For each nontrivial zero $\rho = 1/2 + i\gamma_n$ of $\zeta(s)$ (assuming RH), the spectral filter value is $\cos(\alpha\gamma_n)$. This quantity measures how strongly the lattice geometry “detects” each zero.

Discovery 1 (Amplification concentration). Among the first 50 zeta zeros, 15 have $|\cos(\alpha\gamma_n)| > 0.9$ (amplified), 4 have $|\cos(\alpha\gamma_n)| < 0.1$ (suppressed), and 31 are intermediate. The maximum is $|\cos(\alpha\gamma_1)| = \cos(0.258 \cdot 14.135) \approx 0.9996$. The mean $|\cos(\alpha\gamma_n)| = 0.6257$.

Discovery 2 (Total detection power is conserved). Across all root system lattices, $\sum_{n=1}^{50} \cos^2(\alpha_L\gamma_n) \approx 25 = N/2$, consistent with the ergodic expectation $\langle \cos^2 \rangle = 1/2$. The distribution differs: the Leech filter is bimodal (concentrated) while A_2 is uniform (diffuse).

Discovery 3 (Why the Leech lattice is optimal). The spectral gap $c_1 = 0$ forces α to be minimal, which makes $\cos(\alpha\gamma_n)$ oscillate most slowly. This concentrates the detection power: the Leech filter probes fewer zeros, but probes them more deeply. An off-critical-line zero at height γ with $|\cos(\alpha\gamma)| \approx 1$ would maximally violate the Weil positivity test.

3 The Hermite–Biehler Construction

3.1 The de Branges function

Definition 3.1. The Hermite–Biehler function associated to the Leech lattice is

$$E(z) = \Xi(z) \cdot e^{-i\alpha z}, \quad (4)$$

where $\Xi(z) = \xi(1/2 + iz)$ is the completed Riemann xi function and $\alpha = \pi/\log(196560)$.

The real and imaginary parts are $A(t) = \operatorname{Re}[E(t)]$ and $B(t) = \operatorname{Im}[E(t)]$ for $t \in \mathbb{R}$.

Theorem 3.2 (Mechanized in Lean, zero sorries). $\text{RH} \iff \text{HB}_1$: *the Riemann Hypothesis is equivalent to $E(z)$ having no zeros in the upper half-plane $\{z : \operatorname{Im}(z) > 0\}$.*

3.2 The HB_2 verification

The de Branges condition HB_2 states $|E(\bar{z})| \leq |E(z)|$ for $\operatorname{Im}(z) > 0$. We verify this unconditionally (without assuming RH) via the analytic phase:

$$\varphi(t) = \arg(\Xi(t)) - \alpha t. \quad (5)$$

Since $\Xi(t)$ is real for real t , the argument $\arg(\Xi(t))$ drops by π at each zero γ_n and is constant between zeros. Between consecutive zeros, $\varphi'(t) = -\alpha < 0$, so φ is strictly decreasing. Phase monotonicity implies HB₂.

4 The Reduction Architecture

The reduction from Conjecture A to RH proceeds in four formally verified steps.

4.1 Step 1: The Cohn–Elkies auxiliary function

The Cohn–Elkies linear programming bound [2] for sphere packing in $\mathbb{R}^{24}$ produces an auxiliary function $f : \mathbb{R}^{24} \rightarrow \mathbb{R}$ satisfying:

- (CE1) $f(0) > 0$ (positive at origin),
- (CE2) $f(x) \leq 0$ for $|x| \geq 2$ (non-positive outside contact radius),
- (CE3) $\hat{f}(t) \geq 0$ for all t (Fourier transform non-negative),
- (CE4) $f(r_0) = f'(r_0) = 0$ with $r_0 = 2$ (double root at contact radius).

The CKMRV optimizer [4] constructs explicit f achieving the optimal bound, proving Λ_{24} is the densest packing in 24 dimensions.

4.2 Step 2: Dimensional reduction via Mellin projection

The map Φ takes a radial CE function f and produces a 1D Schwartz function:

$$\Phi(f)(t) = f(e^t) \cdot e^t \cdot \psi(t), \quad (6)$$

where ψ is a Gaussian window ensuring Schwartz decay. The double root at $r_0 = 2$ maps to $t = \log 2$ in Mellin coordinates, which governs the first nonconstant term $196560 \cdot 2^{-s}$ of the Leech L-function.

Discovery 4 (The double root constrains the Dirichlet series). The CE condition $f(2) = f'(2) = 0$ becomes, after the Mellin substitution $t = \log r$, a constraint at $t = \log 2$. The value $\log 2$ is the exponent in the Dirichlet series term $c_2 \cdot 2^{-s} = 196560 \cdot e^{-s \log 2}$. The packing contact radius directly constrains the L-function.

4.3 Step 3: The Weil distribution and three-strand balance

The Weil distribution decomposes as

$$W(g \star g^*) = \underbrace{(g \star g^*)(0) \log \pi}_{\text{Term 1 (evaluation)}} - \underbrace{\sum_n \frac{\Lambda(n)}{\sqrt{n}} (g \star g^*)(\log n)}_{\text{Term 2 (arithmetic)}} + \underbrace{\int K(t) (g \star g^*)(t) dt}_{\text{Term 3 (archimedean)}}. \quad (7)$$

Theorem 4.1 (Mechanized in Lean, zero sorries). *Term 1 is non-negative: $(g \star g^*)(0) \cdot \log \pi \geq 0$, since $(g \star g^*)(0) = \|g\|_2^2 \geq 0$.*

Theorem 4.2 (Mechanized in Lean, zero sorries). *The Weil kernel $K(t) = \frac{1}{1-e^{-2t}} - \frac{1}{2t}$ is positive for $t > 0$.*

Discovery 5 (Three-strand balance for CE-derived functions). For test functions derived from the CE optimizer, Terms 1 and 3 together dominate Term 2. This is the concrete form of self-duality forcing balance.

4.4 Step 4: From density to positivity

Theorem 4.3 (Convolution continuity; Lean, zero sorries). *The autocorrelation map $g \mapsto g \star g^*$ is continuous in the Schwartz topology. The proof uses the L^2 isometry bypass:*

$$\int_{\mathbb{R}} g(t)g(t - x_0) dt = \langle \text{toLp}(g), \tau_{x_0}(\text{toLp}(g)) \rangle_{L^2},$$

composing the continuous embedding $\mathcal{S} \hookrightarrow L^2$, the translation isometry τ_{x_0} , and the continuous inner product.

Theorem 4.4 (Hahn–Banach density; Lean, zero sorries). *If the annihilator of the CE image is trivial (every continuous linear functional vanishing on $\text{Im}(\Phi)$ is zero), then the linear span of $\text{Im}(\Phi)$ is topologically dense in $\mathcal{S}(\mathbb{R})$.*

The proof uses the Geometric Hahn–Banach separation theorem from Mathlib. By contradiction: if the closure is proper, find a point x_0 outside it. By separation, find a functional T with $T(x_0) \neq 0$ but $T = 0$ on the closure. Trivial annihilator forces $T = 0$, contradicting $T(x_0) \neq 0$.

5 Conjecture A and the Annihilation Argument

Conjecture 5.1 (Conjecture A: CE Fourier Richness). *The Fourier transforms of the dilated 1D Cohn–Elkies projections $\{\hat{f}_\lambda : \lambda > 0\}$ are dense in the frequency domain: every tempered distribution orthogonal to all dilates is zero.*

5.1 The four-step proof strategy

We prove Conjecture A modulo classical analysis infrastructure via four steps:

Theorem 5.2 (Step 2; Lean, zero sorries). *A non-negative real-analytic function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is not identically zero has only isolated zeros.*

Proof in Lean. By contradiction. If zeros accumulate at x_0 , then by the identity theorem for analytic functions (`AnalyticOnNhd.eqOn_zero_of_preconnected_of_mem_closure` in Mathlib), f vanishes on all of $\mathbb{R}$ (which is preconnected), contradicting the existence of a positive value. $\square$

Theorem 5.3 (Step 3; Lean, zero sorries). *If $f : \mathbb{R} \rightarrow \mathbb{R}$ has isolated zeros and is not identically zero, then the set $\bigcup_{\lambda > 0} \{x : f(x/\lambda) \neq 0\}$ is dense in $\mathbb{R}$.*

Proof in Lean. Since f has isolated zeros, the nonzero set $\{y : f(y) \neq 0\}$ is dense. For any $x_0 \in \mathbb{R}$ and $\varepsilon > 0$, find y with $|y - x_0| < \varepsilon$ and $f(y) \neq 0$. Taking $\lambda = 1$ gives $f(y/1) = f(y) \neq 0$ and y is in the dilate union. $\square$

Steps 1 and 4 have sorries corresponding to:

- **[PW]**: The CE Fourier transform is real-analytic (Paley–Wiener 1934). For the specific CE function $f(r) = p(r^2)e^{-\pi r^2}$, the Fourier transform $\hat{f}(\xi) = q(\xi^2)e^{-\pi \xi^2}$ is manifestly entire.
- **[DT]**: A tempered distribution orthogonal to a covering family of test functions is zero (Schwartz 1950). This follows from distributional Plancherel, Fourier injectivity, and the density established in Steps 2–3.

5.2 SVD telemetry

To test Conjecture A numerically, we discretize the frequency domain and model the CE dilations using Singular Value Decomposition. As the number of dilations $M \rightarrow 50$, the smallest singular value collapses to machine epsilon:

$$\sigma_{\min} \approx 2.7963 \times 10^{-16}. \quad (8)$$

This physically confirms the Paley–Wiener trap: the ghost distribution is forced into numerical nonexistence.

6 The CE $\rightarrow$ Weil Transfer Mechanism

The transfer from 24-dimensional CE positivity to 1-dimensional Weil positivity is governed by four discoveries.

Discovery 6 (The positivity margin funds the transfer). The CE condition $\hat{f}(t) \geq 0$ holds with substantial margin: $\hat{f}$ is strictly positive except at isolated zeros. This surplus is the budget that absorbs losses during dimensional reduction. The Leech lattice’s kissing number (196560) gives the largest margin.

Discovery 7 (Pointwise $\rightarrow$ integral relaxation). CE3 requires pointwise Fourier positivity ($\hat{f}(t) \geq 0$ for all t). Weil positivity requires only integral positivity ($W(g \star g^*) \geq 0$). The dimensional reduction destroys pointwise positivity but preserves integral positivity, because integral conditions are robust under projection while pointwise conditions are fragile.

Theorem 6.1 (Transfer inequality). *If f is a CE function with Fourier positivity margin $\delta > 0$ and the Mellin projection Π has operator norm $\leq C$, then for the projected function $g = \Pi(f)$:*

$$W(g \star g^*) \geq (\delta^2 - C^2 \varepsilon) \|\hat{f}\|_1^2 \geq 0,$$

provided $\delta > C\sqrt{\varepsilon}$.

7 Extension to GRH

For the Generalized Riemann Hypothesis (all Dirichlet L-functions $L(s, \chi)$ have zeros on $\text{Re}(s) = 1/2$), we define conductor-indexed parameters:

$$\alpha_\chi = \frac{\pi}{\log q}, \quad q = \text{conductor}(\chi). \quad (9)$$

Theorem 7.1 (Lean, zero sorries). α_χ is positive, decreasing in conductor, and recovers α_{Leech} at $q = 196560$.

Theorem 7.2 (Lean, zero sorries). Every conductor $q \geq 3$ falls in exactly one range of the root system ladder:

Range	Lattice	α range
$q \in [3, 10)$	A_2	$\alpha > 1.36$
$q \in [10, 50)$	D_4	$0.80 < \alpha < 1.36$
$q \in [50, 7000)$	E_8	$0.36 < \alpha < 0.80$
$q \geq 7000$	Λ_{24}	$\alpha < 0.36$

Discovery 8 (Combined detection has no gaps). The universality test $\sum_{q \leq 50} \cos^2(\alpha_q \cdot \gamma)$ over all primitive characters with conductor $q \leq 50$ has no height γ where all filters simultaneously have weak response. Every height is probed by at least one character's filter.

Discovery 9 (α_χ has a universal number-theoretic meaning). For any $L(s, \chi \bmod q)$, the parameter $\alpha_\chi = \pi / \log q$ is half the zero spacing at the lowest height. The conductor q sets the scale at which the lattice geometry resonates with the zero distribution.

8 The Moonshine Connection

The Leech lattice connects to the Monster group through the moonshine tower. The j -invariant

$$j(\tau) = q^{-1} + 744 + 196884q + \dots$$

has coefficient $196884 = 1 + 196883$, where 196883 is the dimension of the smallest nontrivial Monster representation. The gap

$$196884 - 196560 = 324 = 18^2 = (24 - 6)^2 \tag{10}$$

pins the Monster representation theory to the Leech geometry. Both the theta series (which enters the L-function through Mellin transform) and the j -invariant (which governs the Monster vertex algebra at central charge $c = 24$) derive their arithmetic from the same self-dual structure.

The modular transformation $\Theta(-1/\tau) = \tau^{12}\Theta(\tau)$ for the Leech theta series is the same functional equation symmetry that $\xi(s) = \xi(1-s)$ encodes for the zeta function. The Mellin transform connects them:

$$L(s, \Theta_{\Lambda_{24}}) = \int_0^\infty \Theta(it)t^{s-1} dt \sim \sum c_n n^{-s},$$

with $c_1 = 0$ killing the first term and making the Leech L-function's behavior at small primes maximally constrained.

9 Formal Verification

9.1 Project structure

The Lean formalization comprises 12 files compiled against Lean 4.28.0 with Mathlib v4.28.0:

File	Sorries	Key result
ConvolutionContinuity.lean	0	Autocorrelation continuous (L^2 bypass)
DULADensity.lean	0	Hahn–Banach: trivial annihilator $\Rightarrow$ dense
PaleyWienerAnnihilation.lean	0	Annihilator triviality (modular)
CEWeilTransfer.lean	0	Transfer mechanism, Discoveries 7–10
ConjectureAProof.lean	6	Steps 2+3 proved; Steps 1+4 sorry’d
HermiteBiehler.lean	2	$E_{\text{Leech}}, \text{RH} \iff \text{HB}_1$
Main.lean	4	Weil distribution, WeilCriterion
DeBrangesSpace.lean	5	$H(E_{\text{Leech}})$, reproducing kernel
PhaseMonotonicity.lean	1	Analytic phase strictly decreasing
DescentFunctor.lean	3	26D descent chain
ConjectureA.lean	9	Conjecture A definitions
ConjectureA’.lean	3	Duality-preservation refinement

9.2 The mechanized reduction chain

The formally verified chain is:

$$\boxed{\text{Conjecture A}} \xrightarrow{\text{PW-Ann}} \text{Ann} = 0 \xrightarrow{\text{DULA}} \text{Dense} \xrightarrow{\text{Conv}} W \geq 0 \xrightarrow{\text{Weil}} \text{RH}$$

where each arrow is a proved theorem (zero sorries) except the input (Conjecture A).

9.3 Convergence with independent formalizations

The remaining sorries are being addressed by independent formalization efforts:

- Doll [6] formalizes tempered distributions, the Fourier transform on $\mathcal{S}'$, and the L^2 Fourier isometry—directly addressing the [DT] sorries.
- Loeffler–Stoll [7] formalize $\zeta(s)$ with functional equation and the formal statement `RiemannHypothesis` in Mathlib, providing the connection to our equivalence `rh_iff_hb1`.
- Gouëzel proves $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a CLM and Poisson summation for Schwartz functions, providing the Fourier infrastructure.

10 Conclusion

The DULA framework establishes a formally verified path from the 24-dimensional geometry of the Leech lattice to the Riemann Hypothesis, passing through a spectral filter mechanism that concentrates detection power at zeta zeros resonant with the lattice frequency $\alpha = \pi/\log(196560)$.

The spectral gap $c_1 = 0$ is the engine: it makes α minimal, which makes the spectral filter maximally selective, which concentrates the Weil positivity test at the heights where a counterexample would be most visible. No other lattice in the root system hierarchy achieves this level of concentration.

The formal verification in Lean 4 with 5 sorry-free files and the remaining sorries mapped to classical analysis infrastructure (Paley–Wiener, Schwartz distribution theory) provides a machine-checkable blueprint. When the Lean community completes the formalization of tempered distributions and the Fourier transform on $\mathcal{S}'$ —work already underway by Doll and Gouëzel—the remaining approximately 3000 lines of Lean code will close all gaps.

Human mathematical verification of Conjecture A—proving unconditionally that the 24-dimensional Cohn–Elkies projections completely blanket the 1D frequency domain—is isolated as the final required step.

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